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Light irradiation device

Abstract

A light irradiation device includes a tube including a tube main body and a light emitting portion, a plurality of wiring boards disposed inside the light emitting portion and including a plurality of light emitting elements disposed at an equal interval mounted, an anode lead wire for connecting an anode terminal of each of the wiring boards to a power source, and a cathode lead wire for connecting a cathode terminal of each of the wiring boards to the power source. In the light emitting portion, a central lumen and surrounding lumens that are arrayed at a 90° interval around the central lumen are formed. The wiring boards are disposed in the respective surrounding lumens with the light emitting elements arrayed along a longitudinal direction of the light emitting portion.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5814039	12/1997	Prescott	N/A	N/A
11229808	12/2021	Barneck	N/A	A61L 2/085
2014/0235942	12/2013	Hellstrom et al.	N/A	N/A
2016/0059031	12/2015	Wescott et al.	N/A	N/A
2018/0008122	12/2017	Arai	N/A	A61B 1/00013
2018/0035912	12/2017	Hayman et al.	N/A	N/A
2019/0217117	12/2018	Barneck	N/A	A61M 25/0028

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
110234397	12/2018	CN	N/A
H08505803	12/1995	JP	N/A
2007511279	12/2006	JP	N/A
2016502419	12/2015	JP	N/A
2020043897	12/2019	JP	N/A
WO2006/130365	12/2005	WO	N/A
WO-2019088940	12/2018	WO	A61N 5/062

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a light irradiation device for irradiating an affected area with light having a specific wavelength required for photoimmunotherapy.

BACKGROUND ART

(2) A recent light irradiation device can perform photoimmunotherapy against cancer relatively easily without large-scale equipment, such as a laser oscillator. An introduced example (light-emitting treatment device) of such a light irradiation device includes a tube provided with a light emitting portion having light permeability on a distal end side, a flexible wiring board disposed inside the light emitting portion in the tube and including a light emitting element that emits light having a specific wavelength mounted, a main body portion that is connected to a proximal end portion of the tube and includes a power source for supplying power to the light emitting element, and a lead wire that is inserted in the tube has one end connected to an electrode of the flexible wiring board and the other end connected to the power source (see Patent Document 1 below).

(3) In performing photoimmunotherapy against bile duct cancer with such a light irradiation device, first, an endoscope (side-viewing endoscope) is inserted through a mouth of a patient until a distal end portion of the endoscope reaches the duodenal papilla. Then, the tube is inserted into a bile duct along a guide wire inserted into the bile duct in advance, and the light emitting portion is disposed in contact with or close to the bile duct cancer. Next, the guide wire and the endoscope are removed, leaving only the tube in the body. Then, a proximal end portion of the tube extending from the mouth is extended from the nostril via the nasal cavity.

CITATION LIST

Patent Literature

(4) Patent Document 1: JP 2020-43897 A

SUMMARY OF INVENTION

Technical Problem

(5) For the photoimmunotherapy against bile duct cancer, the entirety of the bile duct cancer developed on the inner wall of the bile duct is preferably irradiated with light uniformly.

(6) Unfortunately, the known light irradiation device has variation in the amount of light emitted (the amount of light irradiation to the cancer) from the light emitting portion.

(7) For example, Patent Document 1 above discloses a light irradiation device in which the flexible wiring board is disposed inside the light emitting portion in the tube. The flexible wiring board is formed in a shape bent at a bent portion with a pair of side portions oppositely disposed and includes a plurality of light emitting elements mounted on each of the side portions. Unfortunately, such a light irradiation device causes variation in the amount of emitted light in a circumferential direction of the light emitting portion.

(8) Patent Document 1 also discloses a light irradiation device in which the flexible wiring board is disposed inside the light emitting portion in the tube. The flexible wiring board is formed twisted into a spiral shape and includes a plurality of light emitting elements mounted on its outer surface. Unfortunately, such a light irradiation device causes variation in the amount of emitted light also in a longitudinal direction of the light emitting portion.

(9) The photoimmunotherapy against bile duct cancer requires the light emitting portion of the light irradiation device to be positioned against the bile duct cancer narrowing the bile duct and the bile duct cancer to be irradiated with light for a long period of time (about two weeks).

(10) However, while the bile duct cancer is irradiated with light, the light emitting portion of the light irradiation device may block the bile duct constricted by the bile duct cancer. This may cause bile to be accumulated in the gallbladder, causing the patient to suffer from jaundice.

(11) The present invention has been made in view of the above-described circumstances.

(12) An object of the present invention is to provide a light irradiation device capable of irradiating cancer developed on an inner wall of a body duct with light having a specific wavelength in a circumferential direction of the body duct uniformly without variation in the amount of light emitted from the light emitting portion in the circumferential direction.

(13) Another object of the present invention is to provide a light irradiation device capable of irradiating the entirety of cancer developed on an inner wall of a body duct with light having a

specific wavelength uniformly without variation in the amount of emitted light in the circumferential direction and the longitudinal direction of the light emitting portion.
(14) Still another object of the present invention is to provide a light irradiation device enabling emission of a body fluid (drainage) during photoimmunotherapy even when a light emitting portion is disposed so as to block the portion constricted by cancer.
Solution to Problem
(15) (1) A light irradiation device according to the present invention is a light irradiation device for irradiating an affected area (cancer) with light having a specific wavelength required for photoimmunotherapy and includes: a tube including a tube main body and a light emitting portion that has light permeability and is connected to a distal end side of the tube main body; a plurality of wiring boards disposed inside the light emitting portion of the tube and including a light emitting element that emits the light having the specific wavelength mounted; an anode lead wire extending inside the tube to connect an anode terminal of each of the wiring boards to a power source; and a cathode lead wire extending inside the tube to connect a cathode terminal of each of the wiring boards to the power source.
(16) In the light emitting portion, a central lumen extending along a center axis of the tube and a plurality of surrounding lumens that are arrayed at an equal angular interval around the central lumen (that is, along a circumferential direction of the light emitting portion) and are closed on a distal end side of the light emitting portion are formed, and the wiring boards are disposed in the respective surrounding lumens with the light emitting element disposed facing an outer side of the light emitting portion in a radial direction.
(17) According to the light irradiation device with such a configuration, photoimmunotherapy against cancer can be performed by inserting the tube into the body, disposing the light emitting portion in contact with or close to the affected area (cancer) to which the photosensitive substance for the photoimmunotherapy is attached, energizing the plurality of wiring boards through the anode lead wire and the cathode lead wire, causing the light emitting element mounted on each of the wiring boards, and irradiating the photosensitive substance and the like attached to the cancer with light.
(18) The wiring board on which the light emitting element is mounted is disposed in each of the plurality of surrounding lumens arrayed at an equal angular interval around the central lumen of the light emitting portion. This allows light having a specific wavelength to be radially emitted without variation in the circumferential direction of the light emitting portion, enabling uniform light irradiation to the affected area (the photosensitive substance and the like attached to the cancer) around the light emitting portion.
(19) In addition, even when the light emitting portion is disposed to block a constricted portion of a body duct through which body fluid, such as bile, flows, flowing the body fluid through the central lumen of the light emitting portion allows the body fluid to be emitted to the outside of the body. Further, flowing the body fluid through the central lumen causes light from the light emitting element mounted on the wiring board disposed in each of the surrounding lumens not to be blocked by the body fluid.
(20) (2) In the light irradiation device according to the present invention, preferably, each of the wiring boards includes a plurality of the light emitting elements disposed at a substantially equal interval and mounted, and the wiring boards are disposed in the respective surrounding lumens with the plurality of the light emitting elements arrayed along a longitudinal direction of the light emitting portion while facing the outer side of the light emitting portion in the radial direction.
(21) According to the light irradiation device with such a configuration, the wiring board is disposed in each of the plurality of surrounding lumens arrayed at an equal angular interval around the central lumen of the light emitting portion with the light emitting elements arrayed along the longitudinal direction of the light emitting portion. This allows light having a specific wavelength to be radially emitted without variation in the circumferential direction and the longitudinal

direction of the light emitting portion, enabling uniform light irradiation to the entirety of the affected area (photosensitive substance and the like attached to the cancer) around the light emitting portion.
(22) (3) In the light irradiation device according to the present invention, preferably, four of the surrounding lumens are formed at a 90° interval around the central lumen.
(23) According to the light irradiation device with such a configuration, the four wiring boards can be arranged along the circumferential direction of the light emitting portion, enabling even and uniform light irradiation to the affected area around the light emitting portion.
(24) (4) In the light irradiation device according to the present invention, preferably, in the tube main body of the tube, a first lumen communicating with the central lumen of the light emitting portion and extending in parallel to the center axis while being displaced in a radial direction of the tube from the center axis of the tube and a second lumen extending in parallel to the first lumen and opening to a proximal end surface of the tube are formed, and the anode lead wire and the cathode lead wire are inserted in the second lumen from each of the surrounding lumens, extend in the second lumen, and extend from an opening in the proximal end surface.
(25) According to the light irradiation device with such a configuration, in the tube main body, the anode lead wire and the cathode lead wire extend in the second lumen, and body fluid of a patient can be flowed through the first lumen. Thus, the body fluid of the patient does not come into contact with the anode lead wire and the cathode lead wire inside the tube main body, and, for example, short circuit between the lead wires and contamination of the lead wires due to the contact with the body fluid can be avoided.
(26) In addition, the first lumen is displaced in the radial direction of the tube from the center axis of the tube. Thus, a cross-sectional area of the second lumen serving as the insertion space for the lead wires can be sufficiently increased, and using the light irradiation device of (7) described below allows the body fluid of the patient flowed through the first lumen to be emitted from a side periphery of the tube main body.
(27) (5) In the light irradiation device according to (4) described above, preferably, the tube includes, between the light emitting portion and the tube main body, a lumen switching portion sealing the anode lead wire and the cathode lead wire and formed with a communication path through which the central lumen and the first lumen communicate with each other.
(28) According to the light irradiation device with such a configuration, the communication path of the lumen switching portion allows the central lumen of the light emitting portion and the first lumen of the tube main body to reliably communicate with each other.
(29) In addition, in the lumen switching portion, the anode lead wire and the cathode lead wire are sealed (fixed by the resin forming the lumen switching portion). This can reliably prevent, in connecting, for example, the proximal end portions of the lead wires to the power source, the lead wires from being pulled in the proximal end direction to cause the distal end portions of the lead wires to be detached from electrode terminals of the wiring board or the lead wires being entangled in the lumen switching portion.
(30) (6) In the light irradiation device according to (4) or (5) described above, preferably, a plurality of the anode lead wires and a plurality of the cathode lead wires inserted in the second lumen are each combined into one lead wire, and each lead wire extends from the opening in the proximal end surface of the tube.
(31) According to the light irradiation device with such a configuration, the anode lead wires and the cathode lead wires are each combined into one lead wire, enabling simple operation of connecting the proximal end portions of these lead wires to the power source.
(32) (7) In the light irradiation device according to (4) to (6) described above, preferably, the first lumen is closed on a proximal end side of the tube, and a side hole extending from an outer periphery of the tube main body to the first lumen is formed in a peripheral wall of the tube main body.

(33) According to the light irradiation device with such a configuration, the body fluid of the patient flowing through the first lumen can be emitted from the opening of the side hole, allowing the anode lead wire and the cathode lead wire extending from the opening in the proximal end surface of the tube to be prevented from coming into contact with the body fluid emitted (using the opening as a liquid emission port).
(34) (8) In the light irradiation device according to (7) described above, preferably, the outer periphery of the tube main body is provided with marking indicating a position of an opening of the side hole, allowing the position of an opening with a small diameter to be easily recognized.
(35) (9) In the light irradiation device according to the present invention, preferably, a lead protection sheath is attached to a proximal end side of the tube, and the lead protection sheath encloses the anode lead wire and the cathode lead wire extending from the opening in the proximal end surface of the tube and is easily detachable from the tube.
(36) According to the light irradiation device with such a configuration, the lead wires (the anode lead wire and the cathode lead wire) extending from the opening in the proximal end surface of the tube can be prevented from being wet with snivel or the like when the proximal end portion of the tube extending from the mouth is extended from the nostril via the nasal cavity. In addition, detaching the lead protection sheath from the tube after extending the proximal end portion of the tube from the nostril of the patient allows the unwet proximal end portions of the lead wires to be connected to the power source.
(37) (10) In the light irradiation device according to the present invention, preferably, the tube includes, on a distal end side of the light emitting portion, a most distal end portion having a single lumen structure formed with a lumen communicating with the central lumen.
(38) According to the light irradiation device with such a configuration, the body fluid on the distal end side of the light emitting portion disposed to block the constricted portion can be flowed through the central lumen of the light emitting portion via the lumen of the most distal end portion and emitted to the outside of the body.
(39) (11) In the light irradiation device according to (10) described above, preferably, a peripheral wall of the most distal end portion includes at least one side hole extending from an outer periphery of the most distal end portion to the lumen.
(40) According to the light irradiation device with such a configuration, even when the body fluid cannot be flowed into the lumen from the opening at the distal end of the most distal end portion due to the opening in contact with the inner wall or the like of the body duct, the body fluid can be flowed into from the side hole.
(41) (12) In the light irradiation device according to (10) or (11) described above, preferably, the lumen of the most distal end portion has a larger diameter than a diameter of the central lumen of the light emitting portion, and the tube includes, between the light emitting portion and the most distal end portion, a lumen diameter expansion portion formed with a diameter expansion communication path through which the central lumen and the lumen of the most distal end portion communicate with each other.
(42) According to the light irradiation device with such a configuration, an amount of inflow of body fluid into the lumen of the most distal end portion and thus an amount of emission of the body fluid can be sufficiently increased.
(43) (13) The light irradiation device according to the present invention can be suitably used for photoimmunotherapy against bile duct cancer or pancreatic cancer.
(44) According to the light irradiation device with such a configuration, bile duct cancer developed on an inner wall of a bile duct or pancreatic cancer developed on an inner wall of a pancreatic duct can be irradiated with light having a specific wavelength uniformly in the circumferential direction of the bile duct and the pancreatic duct. Even when the light emitting portion is disposed to block the portion constricted by the bile duct cancer or the pancreatic cancer, the bile can be emitted.
Advantageous Effects of Invention

(45) The light irradiation device according to the present invention is capable of irradiating a cancer developed on an inner wall of a body duct with light having a specific wavelength in the circumferential direction of the body duct uniformly without variation in the amount of light emitted from the light emitting portion in the circumferential direction.
(46) Each of the wiring boards includes the plurality of light emitting elements disposed at a substantially equal interval and mounted, and the wiring boards are disposed in the respective surrounding lumens with the plurality of light emitting elements arrayed along a longitudinal direction of the light emitting portion while facing the outer side of the light emitting portion in the radial direction. This can irradiate the entirety of the cancer developed on an inner wall of a body duct with light having a specific wavelength uniformly without variation in the amount of emitted light along the circumferential direction and the longitudinal direction of the light emitting portion.
(47) Even when the light emitting portion is disposed to block a constricted portion of the body duct, flowing the body fluid through the central lumen of the light emitting portion enables emission of the body fluid (drainage).
(48) Further, flowing the body fluid through the central lumen causes light from the light emitting element mounted on the wiring boards not to be blocked by the body fluid, nor is the amount of light emitted from the light emitting portion reduced due to the blocking of the body fluid.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** is a plan view illustrating an embodiment of a light irradiation device according to the present invention.
(2) FIG. **2**A is an end surface view taken along IIA-IIA in FIG. **1** (an end surface view of a most distal end portion).
(3) FIG. **2**B is an end surface view taken along IIB-IIB in FIG. **1** (an end surface view of a light emitting portion).
(4) FIG. **2**C is an end surface view taken along IIC-IIC in FIG. **1** (an end surface view of the light emitting portion).
(5) FIG. **2**D is an end surface view taken along IID-IID in FIG. **1** (an end surface view of a tube main body).
(6) FIG. **2**E is an end surface view taken along IIE-IIE in FIG. **1** (an end surface view of the tube main body and a lead protection sheath).
(7) FIG. **2**F is an end surface view taken along IIF-IIF in FIG. **1** (an end surface view of the lead protection sheath).
(8) FIG. **2**G is a detail view of part IIG in FIG. **1**.
(9) FIG. **3** is a perspective view illustrating a state where lead wires extending from each of surrounding lumens of the light emitting portion are inserted in a second lumen of the tube main body in the light irradiation device illustrated in FIG. **1**.
(10) FIG. **4**A is a perspective view illustrating the state where lead wires extending from each of surrounding lumens of the light emitting portion are inserted in the second lumen of the tube main body in the light irradiation device illustrated in FIG. **1**, as viewed from a proximal end side.
(11) FIG. **4**B is a perspective view illustrating the state where lead wires extending from each of surrounding lumens of the light emitting portion are inserted in the second lumen of the tube main body in the light irradiation device illustrated in FIG. **1**, as viewed from a distal end side.
(12) FIG. **5** is a schematic view of the light irradiation device illustrated in FIG. **1**.
(13) FIG. **6** is a schematic view of a power supply device to which the lead wires of the light irradiation device are connected.
(14) FIG. **7**A is a plan view of a wiring board constituting the light irradiation device illustrated in

FIG. **1**.
(15) FIG. **7**B is a side view of the wiring board constituting the light irradiation device illustrated in FIG. **1**.
DESCRIPTION OF EMBODIMENTS
Embodiments
(16) A light irradiation device **100** of the present embodiment illustrated in FIGS. **1** to **5** is a light irradiation device for irradiating bile duct cancer with near infrared light required for photoimmunotherapy against the bile duct cancer and includes: a tube **10** including a tube main body **11**, a light emitting portion **13** that has light permeability and is connected to a distal end side of the tube main body **11** via a lumen switching portion **12**, and a most distal end portion **15** connected to a distal end side of the light emitting portion **13** via a lumen diameter expansion portion **14**; four wiring boards **30** disposed inside the light emitting portion **13** of the tube **10** and including seven light emitting elements **35** that emit near infrared light disposed at an equal interval and mounted; four anode lead wires **51** extending inside the tube **10** to connect respective anode terminals **36** of the wiring boards **30** to a power source; four cathode lead wires **52** extending inside the tube **10** to connect respective cathode terminals **37** of the wiring boards **30** to the power source; and a lead protection sheath **70** attached to a proximal end side of the tube **10**.
(17) In the light emitting portion **13** of the tube **10**, a central lumen **131** extending along a center axis of the tube **10** and surrounding lumens **132** to **135** that are arrayed at a 90° interval around the central lumen **131** and are closed on both end sides of the light emitting portion **13** are formed.
(18) In the tube main body **11** of the tube **10**, a first lumen **111** communicating with the central lumen **131** formed in the light emitting portion **13** and extending while being displaced in a radial direction of the tube **10** from the center axis of the tube **10** and a second lumen **112** extending in parallel to the first lumen **111** are formed, the first lumen **111** is closed on a proximal end side of the tube main body **11**, and a side hole **113** extending from an outer periphery of the tube main body **11** to the first lumen **111** is formed in a peripheral wall of the tube main body **11**, and the second lumen **112** is closed on the distal end side of the tube main body **11** and has an opening **114** in a proximal end surface of the tube **10**.
(19) In the most distal end portion **15** of the tube **10**, a lumen **151** having a larger diameter than a diameter of the central lumen **131** of the light emitting portion **13** and communicating with the central lumen **131** through a diameter expansion communication path **141** of the lumen diameter expansion portion **14** is formed, the lumen **151** of the most distal end portion **15** has an opening **152** in a distal end surface of the tube **10**, and in a peripheral wall of the most distal end portion **15**, a plurality of side holes **153** extending from an outer periphery of the most distal end portion **15** to the lumen **151** is formed.
(20) The four wiring boards **30** are disposed in the respective surrounding lumens **132** to **135** with the light emitting elements **35** arrayed along a longitudinal direction of the light emitting portion **13** while facing an outer side of the light emitting portion **13** in a radial direction.
(21) The four anode lead wires **51** pass through the inside of the lumen switching portion **12** from the respective surrounding lumens **132** to **135** of the light emitting portion **13**, are inserted in the second lumen **112** of the tube main body **11**, extend in the second lumen **112**, and are combined into one combined lead wire **51**G in the middle of the second lumen **112**, and the combined lead wire **51**G extends from the tube **10** through the opening **114**, and the four cathode lead wires **52** also pass through the inside of the lumen switching portion **12** from the respective surrounding lumens **132** to **135** of the light emitting portion **13**, are inserted in the second lumen **112** of the tube main body **11**, extend in the second lumen **112**, and are combined into one combined lead wire **52**G in the middle of the second lumen **112**, and the combined lead wire **52**G extends from the tube **10** through the opening **114**.
(22) The lead protection sheath **70** is attached to enclose the anode lead wire **51**G and the cathode lead wire **52**G extending from the opening **114** in the proximal end surface of the tube **10**.

(23) The light irradiation device **100** of the present embodiment is used for photoimmunotherapy in which a treatment is performed by bringing the light emitting portion **13** into close contact with bile duct cancer developed on the inner wall of the bile duct and irradiating a photosensitive substance selectively attached to the cancerous tissue with photosensitive light from the light emitting elements **35**. Examples of the photosensitive substance can include IR700.
(24) The light irradiation device **100** according to this embodiment includes the tube **10**, the four wiring boards **30**, the four anode lead wires **51** (one combined lead wire **51**G obtained by combining them), the four cathode lead wires **52** (one combined lead wire **52**G obtained by combining them), and the lead protection sheath **70**.
(25) As illustrated in FIGS. **1** and **5**, the tube **10** constituting the light irradiation device **100** includes the tube main body **11**, the lumen switching portion **12**, the light emitting portion **13**, the lumen diameter expansion portion **14**, and the most distal end portion **15** in this order from the proximal end side to the distal end side.
(26) The outer diameter of the tube **10** typically ranges from 1.5 to 15 mm and is 2.3 mm as a preferable example.
(27) The length (effective length) of the tube **10** typically ranges from 300 to 5000 mm and is 1500 mm as a preferable example.
(28) The light emitting portion **13** of the tube **10** has light permeability. In the light emitting portion **13**, the central lumen **131** and four surrounding lumens **132** to **135** arrayed at a 90° interval along the circumferential direction of the light emitting portion **13** around the central lumen **131** are formed.
(29) As illustrated in FIGS. **2**B, **2**C, **3** and **4**A, the central lumen **131** has a circular cross-section and extends along the center axis of the tube **10**.
(30) The central lumen **131** is a bile flowing lumen of the light emitting portion **13** and is open on the distal end side and the proximal end side of the light emitting portion **13**.
(31) The diameter of the central lumen **131** typically ranges from 0.1 to 13 mm and is 0.7 mm as a preferable example.
(32) The surrounding lumens **132** to **135** have a rectangular cross section with the four corners rounded (curved). The surrounding lumens **132** to **135** are lumens for accommodating the wiring boards **30**, are closed by resin constituting the lumen switching portion **12** on the proximal end side of the light emitting portion **13**, and are closed by resin constituting the lumen diameter expansion portion **14** on the distal end side of the light emitting portion **13**.
(33) The size (length×width) of the cross section of each of the surrounding lumens **132** to **135** typically ranges from (0.3 mm×0.3 mm) to (3 mm×3 mm) and is (0.6 mm×0.9 mm) as a preferable example.
(34) The length of the light emitting portion **13** typically ranges from 3 to 500 mm and is 40 mm as a preferable example.
(35) The light emitting portion **13** is formed by a material having light permeability. Examples of such a material can include transparent resin materials such as epoxy resin, polycarbonate, acrylic resin, ABS resin, silicone resin, and urethane resin.
(36) The tube main body **11** of the tube **10** is connected to the proximal end side of the light emitting portion **13** via the lumen switching portion **12**.
(37) The first lumen **111** and the second lumen **112** are formed in the tube main body **11**.
(38) As illustrated in FIGS. **2**D, **2**E, **3** and **4**B, the first lumen **111** has a circular cross-section, is displaced from the center axis of the tube **10** in the radial direction, and extends in parallel to the center axis.
(39) The first lumen **111** is a bile flowing lumen of the tube main body **11**, is open on the distal end side of the tube main body **11**, and communicates with the central lumen **131** of the light emitting portion **13** through a communication path **121** formed in the lumen switching portion **12**.
(40) The diameter of the first lumen **111** typically ranges from 0.4 to 13 mm and is 0.85 mm as a

preferable example.
(41) The first lumen **111** is closed on the proximal end side of the tube main body **11**, and thus the bile flowing through the first lumen **111** is not emitted from the proximal end surface of the tube **10**.
(42) Thus, a side hole **113** extending from the outer periphery of the tube main body **11** to the first lumen **111** is formed in the peripheral wall of the tube main body **11**, to serve as an emission path for the bile flowing through the first lumen **111**. An opening of the side hole **113** in the outer periphery of the tube main body **11** serves as an emission port for the bile flowing through the first lumen **111**.
(43) The shape of the opening of the side hole **113** is, for example, a capsule shape (oval shape), and the size (minor diameter×major diameter) of the opening is 0.8 mm×10 mm as a preferable example.
(44) Preferably, the outer periphery of the tube main body **11** is provided with marking indicating the position of the opening such as painting the edge of the opening, allowing an operator to easily recognize the position of such an opening with a small diameter.
(45) The second lumen **112** has a capsule-shaped (oval-shaped) cross section and is displaced from the center axis of the tube **10** in the radial direction (in a direction opposite to the direction in which the first lumen **111** is displaced) and extends parallel to the center axis and the first lumen **111**.
(46) The second lumen **112** is closed by the resin constituting the lumen switching portion **12** on the distal end side of the tube main body **11**.
(47) The second lumen **112** is open on the proximal end side of the tube main body **11**, and has the opening **114** in the proximal end surface of the tube **10** (tube main body **11**).
(48) The second lumen **112** is a lumen through which the lead wires (the anode lead wires **51**, the combined lead wire **51**G, the cathode lead wires **52**, and the combined lead wire **52**G) are inserted.
(49) The size (minor diameter×major diameter) of the cross section of the second lumen **112** typically ranges from (0.3 mm×0.3 mm) to (5 mm×5 mm) and is (0.7 mm×1.1 mm) as a preferable example.
(50) The length of the tube main body **11** typically ranges from 500 to 4960 mm and is 1400 mm as a preferable example.
(51) The tube main body **11** is formed by resin. Examples of this resin material can include synthetic resin such as PEBAX, polyurethane, polyamide, polyethylene, and polypropylene.
(52) The lumen switching portion **12** of the tube **10** is interposed between the light emitting portion **13** and the tube main body **11**.
(53) In the lumen switching portion **12**, the communication path **121** through which the central lumen **131** of the light emitting portion **13** and the first lumen **111** of the tube main body **11** communicate with each other is formed as a flow path for the bile.
(54) The length of the lumen switching portion **12** typically ranges from 1 to 100 mm and is 10 mm as a preferable example.
(55) The lumen switching portion **12** is formed by resin, and the proximal ends of the surrounding lumens **132** to **135** of the light emitting portion **13** and the distal end of the second lumen **112** of the tube main body **11** are closed by the resin constituting the lumen switching portion **12**.
(56) Examples of the resin material constituting the lumen switching portion **12** can include the resin constituting the tube main body **11** described above as an example.
(57) The most distal end portion **15** of the tube **10** is connected to the distal end side of the light emitting portion **13** via the lumen diameter expansion portion **14**.
(58) The lumen **151** is formed in the most distal end portion **15**.
(59) As illustrated in FIG. **2**A, the lumen **151** has a circular cross section and extends along the center axis of the tube **10**.
(60) The lumen **151** is open on the proximal end side of the most distal end portion **15**, and communicates with the central lumen **131** of the light emitting portion **13** through the diameter

expansion communication path **141** formed in the lumen diameter expansion portion **14**.
(61) The lumen **151** is also open on the distal end side of the most distal end portion **15**, and has the opening **152** at the distal end of the tube **10**.
(62) The plurality of side holes **153** extending from the outer periphery of the most distal end portion **15** to the lumen **151** are formed in the peripheral wall of the most distal end portion **15**.
(63) The most distal end portion **15** of the tube **10** is a drainage tube through which the bile accumulated on the distal end side of the light emitting portion **13** flows into the lumen **151** to be emitted to the outside of the body, and the opening **152** and the opening of the side hole **153** serve as inflow ports of the bile.
(64) The diameter the lumen **151** is larger than that of the central lumen **131**, typically ranges from 0.5 to 14 mm, and is 1.7 mm as a preferable example.
(65) This can sufficiently increase the amount of inflow of bile into the lumen **151** and thus the amount of emission of the bile.
(66) FIG. **1** illustrates the most distal end portion **15** in a shape curved in J which is its remembered shape. This curved shape can be easily deformed through application of external force (deformed into a linear shape for example), but the curved shape is restored once the external force is released. The most distal end portion **15** of such a curved shape can be latched to the inner wall of the bile duct to dispose the light irradiation device **100** at a predetermined position.
(67) It is a matter of course that the mode for facilitating the latching of the most distal end portion (drainage tube) is not limited to this.
(68) The length of the most distal end portion **15** typically ranges from 10 to 100 mm and is 40 mm as a preferable example.
(69) The lumen diameter expansion portion **14** of the tube **10** is interposed between the light emitting portion **13** and the most distal end portion **15**.
(70) In the lumen diameter expansion portion **14**, the diameter expansion communication path **141** through which the central lumen **131** of the light emitting portion **13** and the lumen **151** of the most distal end portion **15** communicate with each other is formed as a flow path for the bile.
(71) The diameter expansion communication path **141** is a space of a circular truncated cone shape. The diameter of the diameter expansion communication path **141** at the proximal end of the lumen diameter expansion portion **14** is the same as the diameter of the central lumen **131**. The diameter of the diameter expansion communication path **141** at the distal end of the lumen diameter expansion portion **14** is the same as the diameter of the lumen **151**.
(72) The length of the diameter expansion communication path **141** typically ranges from 1 to 30 mm and is 10 mm as a preferable example.
(73) The lumen diameter expansion portion **14** is formed by resin, and the distal ends of the surrounding lumens **132** to **135** of the light emitting portion **13** are closed by the resin constituting the lumen diameter expansion portion **14**.
(74) Examples of the resin material constituting the lumen diameter expansion portion **14** can include the resin constituting the tube main body **11** described above as an example.
(75) When the light emitting portion **13** is disposed to block the constricted portion of the bile duct, the bile of the patient accumulated around the most distal end portion **15** flows into the lumen **151** from the opening **152** at the distal end of the most distal end portion **15** and the opening of the side hole **153**.
(76) The bile that has flowed into the lumen **151** of the most distal end portion **15** flows in the lumen **151**, the diameter expansion communication path **141** of the lumen diameter expansion portion **14**, the central lumen **131** of the light emitting portion **13**, the communication path **121** of the lumen switching portion **12**, and the first lumen **111** of the tube main body **11**, and is emitted to the outside of the body through the opening (emission port) of the side hole **113** formed in the peripheral wall of the tube main body **11**.
(77) As illustrated in FIGS. **2**B, **2**C, **3**, and **4**A, the wiring board **30** is disposed in each of the

surrounding lumens **132** to **135** of the light emitting portion **13** of the tube **10**.
(78) As illustrated in FIGS. **7**A and **7**B, the wiring board **30** constituting the light irradiation device **100** has a thin elongated plate shape.
(79) On one surface of the wiring board **30**, the seven light emitting elements **35** consisting of LEDs that emit near infrared light (for example, wavelengths of 680 to 700 nm) are disposed at an equal interval along the longitudinal direction of the wiring board **30** and mounted.
(80) The anode terminal **36** and the cathode terminal **37** are disposed on the proximal end side of the wiring board **30**.
(81) The length of the wiring board **30** typically ranges from 3 to 450 mm and is 40 mm as a preferable example.
(82) The width of the wiring board **30** typically ranges from 0.3 to 12.5 mm and is 0.55 mm as a preferable example.
(83) The length (L**35** in FIG. **7**A) of the mounting region of the light emitting elements **35** typically ranges from 5 to 455 mm and is 30 mm as a preferable example.
(84) The wiring board **30** is disposed in each of the surrounding lumens **132** to **135** with the light elements **35** mounted thereon arrayed along the longitudinal direction of the light emitting portion **13** while facing the outer side in the radial direction of the light emitting portion **13**.
(85) This causes the four wiring boards **30** to be disposed at a 90° interval along the circumferential direction of the light emitting portion **13**. On each of these wiring boards **30**, the seven light emitting elements **35** arrayed at an equal interval along the longitudinal direction of the light emitting portion **13**, thus the near infrared light from the light emitting elements **35** can be emitted radially in the circumferential direction and the longitudinal direction of the light emitting portion **13** without variation, and the entirety of the bile duct cancer (the photosensitive substance or the like attached thereto) around the light emitting portion **13** can be uniformly irradiated with the near infrared light.
(86) Each of the surrounding lumens **132** to **135** in which the wiring boards **30** are disposed may be filled with a transparent resin material (the resin that is the same as that constituting the light emitting portion for example), to bury the wiring boards **30** in the light emitting portion **13**.
(87) This configuration can prevent the displacement of the wiring board **30**, light scattering, and the like.
(88) As illustrated in FIGS. **3**, **4**A and **5**, the anode terminal **36** and the cathode terminal **37** of the wiring board **30** are respectively connected with the distal end portion of the anode lead wire **51** and the distal end portion of the cathode lead wire **52**. Note that FIG. **5** illustrates only two of the four wiring boards **30**.
(89) The four anode lead wires **51** of the light irradiation device **100** passes through the inside of the lumen switching portion **12** from the respective surrounding lumens **132** to **135** of the light emitting portion **13** accommodating the wiring boards **30** including the anode terminals **36** to which their distal end portions are connected, to be inserted into the second lumen **112** of the tube main body **11**, extend in the second lumen **112**, and are combined into the one combined lead wire **51**G in the middle of the second lumen **112**. The combined lead wire **51**G extends from the tube **10** from the opening **114**.
(90) Similarly to the anode lead wires **51**, the four cathode lead wires **52** of the light irradiation device **100** passes through the inside of the lumen switching portion **12** from the respective surrounding lumens **132** to **135** of the light emitting portion **13** accommodating the wiring boards including the cathode terminals **37** to which their distal end portions are connected, to be inserted into the second lumen **112** of the tube main body **11**, extend in the second lumen **112**, and are combined into the one combined lead wire **52**G in the middle of the second lumen **112**. The combined lead wire **52**G extends from the tube **10** from the opening **114**.
(91) According to the configuration in which the four anode lead wires **51** are combined into the one combined lead wire **51**G, the four cathode lead wires **52** are combined into the one combined

lead wire **52**G, and the combined lead wire **51**G and the combined lead wire **52**G extend out from the tube **10** as described above, an operation of connecting to the power source only needs to be performed once on each of the anode side and the cathode side and thus can be simplified.
(92) As illustrated in FIGS. **3**, **4**A, **4**B, and **5**, the four anode lead wires **51** and the four cathode lead wires **52** are fixed (sealed) in the lumen switching portion **12** by using the resin material constituting the lumen switching portion **12**.
(93) Such a configuration can reliably prevent, in connecting the proximal end portion of the combined lead wire **51**G or the combined lead wire **52**G to the power source, the lead wire from being pulled in the proximal end direction to cause the distal end portion of the anode lead wire **51** or the cathode lead wire **52** to be detached from the anode terminal **36** or the cathode terminal **37** of the wiring board **30** or a plurality of lead wires (the anode lead wires **51** and/or the cathode lead wires **52**) from being entangled in the lumen switching portion.
(94) As illustrated in FIG. **5**, the proximal end portion of the combined lead wire **51**G and the proximal end portion of the combined lead wire **52**G are each connected to a device-side connector **55**.
(95) When the device-side connector **55** is connected to a power source-side connector **95** of a power supply device **90** illustrated in FIG. **6**, the proximal end portion of the combined lead wire **51**G and the proximal end portion of the combined lead wire **52**G are each connected to a power source **91** (anode and cathode) of the power supply device **90**.
(96) This causes the anode terminal **36** of each of the wiring boards **30** to be electrically connected to the power source **91** (anode) of the power supply device **90** via the anode lead wire **51** (combined lead wire **51**G) and the cathode terminal **37** of each of the wiring boards **30** to be electrically connected to the power source **91** (cathode) of the power supply device **90** via the anode lead wire **52** (combined lead wire **52**G).
(97) The power source **91** constituting the power supply device **90** is not limited to a particular source and consists of, for example, a button battery.
(98) The power supply device **90** includes an ON-OFF switch **93** for the power source **91** and a variable resistor **97** with which current from the power source **91** varies for the adjustment of the amount of light emitted from the light emitting elements **35**.
(99) As illustrated in FIGS. **1**, **2**E, **2**F, and **5**, the lead protection sheath **70** is attached to the proximal end side of the tube **10**.
(100) The lead protection sheath **70** of the light irradiation device **100** is attached to completely enclose the combined lead wire **51**G and the combined lead wire **52**G extending from the opening **114** in the proximal end surface of the tube **10**.
(101) The lead protection sheath **70** consists of a peel-away sheath having a proximal end portion provided with a cut **75**, and can be torn from the cut **75** to be easily removable from the tube **10**.
(102) The length of the lead protection sheath **70** typically ranges from 10 to 3000 mm and is 1100 mm as a preferable example.
(103) In this embodiment, the lead protection sheath **70** is attached to the proximal end side of the tube **10**, with a distal end portion **71** of the lead protection sheath **70** overlapped with and tentatively bonded to the proximal end portion of the tube **10** (for example, 60 to 100 mm from the proximal end and 80 mm from the proximal end) as a preferable example.
(104) Photoimmunotherapy against bile duct cancer can be performed by using the light irradiation device **100** of the present embodiment as in (1) to (8) below. (1) Conjugates of antibodies and photosensitive substances (for example, IR700) are administered into the body by intravenous injection or the like, and the conjugates are attached to the cancerous tissue of the bile duct cancer. (2) The tube **10** of the light irradiation device **100** is inserted into a working lumen of an endoscope (side-viewing endoscope). (3) The endoscope is inserted into the mouth of the patient until the distal end portion of the endoscope reaches the duodenal papilla. (4) A guide wire inserted through the inside of the tube **10** (the first lumen **111**, the communication path **121**, the central lumen **131**,

the diameter expansion communication path **141**, and the lumen **151**) is inserted into the bile duct. (5) The tube **10** is inserted into the bile duct along the guide wire, and the light emitting portion **30** is disposed in contact with the bile duct cancer (that is, so as to block the portion constricted by the bile duct cancer with the light emitting portion **30**). (6) The guide wire and the endoscope are removed, leaving only the tube **10** in the body. (7) The proximal end portion of the tube **10** extending from the mouth is extended from the nostril via the nasal cavity. (8) A switch **93** of the power supply device **90** is turned on to supply power from the power source **91** to the light emitting elements **35** of the wiring boards **30**, to make the light emitting elements **35** emit light with which the bile duct cancer around the light emitting portion **13** is irradiated. This can perform the photoimmunotherapy through reaction of the photosensitive substance attached to the cancerous tissue.
(105) In the light irradiation device **100** of this embodiment, the wiring boards **30** each having the seven light emitting elements **35** arrayed at an equal interval along the longitudinal direction of the light emitting portion **13** are disposed in the four respective surrounding lumens **132** to **135** arrayed at a 90° interval around the central lumen **131** of the light emitting portion **13**. Thus, the near infrared light can be radially emitted without variation in the circumferential direction and the longitudinal direction of the light emitting portion **13**, enabling uniform light irradiation to the entirety of bile duct cancer around the light emitting portion **13**.
(106) Even when the light emitting portion **13** is arranged to block the constricted portion of the bile duct, the bile can be flowed into the lumen **151** of the most distal end portion **15** serving as the drainage tube and flowed through the diameter expansion communication path **141** of the lumen diameter expansion portion **14**, the central lumen **131** of the light emitting portion **13**, the communication path **121** of the lumen switching portion **12**, and the first lumen **111** of the tube main body **11**, and emitted to the outside of the body through the opening of the side hole **113** formed in the peripheral wall of the tube main body **11**.
(107) The bile flowing through the central lumen **13** does not block the near infrared light emitted from the light emitting elements **35** mounted on the wiring boards **30** disposed in the respective surrounding lumens **132** to **135**.
(108) In the tube main body **11**, the anode lead wires **51** (combined lead wire **51**G) and the cathode lead wires **52** (combined lead wire **52**G) extend in the second lumen **112**, and the bile of the patient can be flowed through the first lumen **111**. Thus, the bile of the patient does not come into contact with these lead wires, and, for example, short circuit between the anode lead wires **51** and the cathode lead wires **52** and contamination of the lead wires due to the contact with the bile can be avoided.
(109) The combined lead wire **51**G and the combined lead wire **52**G extend from the opening **114** of the second lumen **112** in the proximal end surface of the tube **10** and the bile flowing through the first lumen **111** can be emitted to the outside of the body through the side hole **113** formed in the peripheral wall of the tube main body **11**. Thus, the combined lead wire **51**G and the combined lead wire **52**G extending from the opening **114** can be prevented from coming into contact with the bile.
(110) The anode lead wires **51** and the cathode lead wires **52** sealed (fixed) in the lumen switching portion **12** by using the resin material constituting the lumen switching portion **12**. This can reliably prevent, in connecting the proximal end portion of the combined lead wire **51**G or the combined lead wire **52**G to the power source, the lead wire from being pulled in the proximal end direction to cause the distal end portion of the anode lead wire **51** or the cathode lead wire **52** to be detached from the anode terminal **36** or the cathode terminal **37** of the wiring board **30** or a plurality of lead wires (the anode lead wires **51** and/or the cathode lead wires **52**) from being entangled in the lumen switching portion.
(111) According to the four anode lead wires **51** combined into the one combined lead wire **51**G and the four cathode lead wires **52** combined into the one combined lead wire **52**G in the second lumen **112** of the tube main body **11** and with the combined lead wire **51**G and the combined lead

wire **52**G extending out from the tube **10**, an operation of connecting to the power source only needs to be performed once on each of the anode side and the cathode side and thus can be simplified.
(112) The lead protection sheath **70** consisting of a peel-away sheath is attached to the proximal end side of the tube main body **11** to enclose the combined lead wire **51**G and the combined lead wire **52**G extending from the opening **114** of the second lumen **112**. Thus, when the proximal end portion of the tube **10** passes through the nasal cavity of the patient to extend from the nostril, the combined lead wires **51**G and **52**G can be prevented from being wet with snivel of the like. Detaching the lead protection sheath **70** after extending the proximal end portion of the tube **10** from the nostril of the patient allows the unwet combined lead wires **51**G and **52**G to be connected to the power source.
(113) While an embodiment according to the present invention is described above, the present invention is not limited to this, and modifications can be made as appropriate.
(114) For example, the number of surrounding lumens in which the wiring boards are disposed in the light emitting portion is not limited to four, and may be three to six.
(115) Still, if the number of wiring boards is less than four (meaning that the arranged angle interval exceeds 90°), there may be a region, in the circumferential direction of the light emitting portion, where the amount of emitted light is insufficient. In view of this, the number of wiring boards is preferably four to six, and is most preferably four.
(116) The cancer to be treated with photoimmunotherapy using the light irradiation device according to the present invention is not limited to bile duct cancer, and the treatment can be applied to pancreatic cancer, lung cancer, esophageal cancer, colon cancer, stomach cancer, bladder cancer, prostate cancer, and the like.
REFERENCE SIGNS LIST
(117) **100** Light irradiation device **10** Tube **11** Tube main body **111** First lumen **112** Second lumen **113** Side hole **114** Opening (proximal end of tube) **12** Lumen switching portion **121** Communication path **13** Light emitting portion **131** Central lumen **132** to **135** Surrounding lumen **14** Lumen diameter expansion portion **141** Diameter expansion communication path **15** Most distal end portion **151** Lumen at most distal end portion **152** Opening (distal end of tube) **153** Side hole **30** Wiring board **35** Light emitting element **36** Anode terminal **37** Cathode terminal **51** Anode lead wire **51**G Combined lead wire **52** Cathode lead wire **52**G Combined lead wire **55** Device-side connector **70** Lead protection sheath **71** Distal end portion of lead protection sheath **75** Cut **90** Power supply device **91** Power source **93** ON-OFF switch **95** Power source-side connector **97** Variable resistor

Claims

1. A light irradiation device comprising: a tube including a light emitting portion and inserted into a body, wherein the light emitting portion includes a central lumen extending along a center axis of the tube, at least one surrounding lumen arranged around the central lumen, and at least one light emitting element disposed inside the at least one surrounding lumen, the central lumen is open on a distal end side of the tube, and the at least one surrounding lumen is closed on the distal end side of the tube.
2. The light irradiation device according to claim 1, wherein the light emitting portion includes a plurality of the surrounding lumens, and in a cross-sectional view perpendicular to a central axis, the plurality of the surrounding lumens is arrayed along a circumferential direction of the central lumen.
3. The light irradiation device according to claim 2, wherein the light emitting portion includes four of the plurality of the surrounding lumens, and in the cross-sectional view perpendicular to the central axis, four of the plurality of the surrounding lumens are arrayed at equal intervals from each

other around the central lumen.

4. The light irradiation device according to claim 1, wherein the at least one surrounding lumen extends along a central axis, the light emitting portion includes a plurality of the light emitting elements, and the plurality of the light emitting elements is arrayed along the central axis.

5. The light irradiation device according to claim 1, wherein the light emitting portion includes a wiring board in the at least one surrounding lumen on which the at least one light emitting element is mounted, the wiring board includes an anode terminal and a cathode terminal, and the light irradiation device includes at least one anode lead wire connected to the anode terminal and at least one cathode lead wire connected to the cathode terminal.

6. The light irradiation device according to claim 5, wherein the tube includes a tube main body on the proximal end side of the tube from the light emitting portion, the tube main body includes a first lumen communicating with the central lumen and extending in parallel to the center axis, and a second lumen extending along the first lumen and opening on a proximal end side of the tube, and the at least one anode lead wire and the at least one cathode lead wire are inserted in the second lumen from the at least one surrounding lumen, pass through the second lumen, and extend from an opening of the second lumen.

7. The light irradiation device according to claim 6, wherein the tube includes, between the light emitting portion and the tube main body, a lumen switching portion, and the lumen switching portion seal the at least one anode lead wire and the at least one cathode lead wire and includes a communication path through which the central lumen and the first lumen communicate with each other.

8. The light irradiation device according to claim 6, wherein the light emitting portion includes a plurality of the surrounding lumens, the plurality of the surrounding lumens each accommodate the wiring board, a plurality of the anode lead wires inserted in the second lumen from each of the plurality of the surrounding lumens are combined with each other and extends from the opening of the second lumen, and a plurality of the cathode lead wires inserted in the second lumen from each of the plurality of the surrounding lumens are combined with each other and extends from the opening of the second lumen.

9. The light irradiation device according to claim 6, wherein the first lumen is closed on a proximal end surface of the tube main body, and the tube main body includes a side hole extending from an outer periphery of the tube main body to the first lumen.

10. The light irradiation device according to claim 9, wherein the tube main body includes a mark on the outer periphery indicating a position of the side hole.

11. The light irradiation device according to claim 6, further comprising: a lead protection sheath enclosing the at least one anode lead wire and the at least one cathode lead wire extending from the opening of the second lumen and detachable from the tube.

12. The light irradiation device according to claim 1, wherein the tube includes a most distal end portion on a distal end side of the tube from the light emitting portion, and the most distal end portion includes a lumen communicating with the outside of the tube and the central lumen.

13. The light irradiation device according to claim 12, wherein the most distal end portion includes at least one side hole extending from an outer periphery of the most distal end portion to the lumen.

14. The light irradiation device according to claim 12, wherein the lumen has a larger diameter than a diameter of the central lumen, and the tube includes a lumen diameter expansion portion between the light emitting portion and the most distal end portion, and the lumen diameter expansion portion includes a diameter expansion communication path through which the central lumen and the lumen communicate with each other.

15. The light irradiation device according to claim 1 to be used for photoimmunotherapy against at least one of bile duct cancer or pancreatic cancer.